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Research article



Integrating fine-tuning and retrieval-augmented generation to address the application challenges of pollution control guidelines

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ABSTRACT

Environmental pollution has become increasingly complex, prompting governmental agencies to issue numerous technical and detailed guidelines for pollution control. While these documents contain substantial domain-specific knowledge, their complexity often impedes effective implementation and broader public comprehension. This study presents the first systematic application of combining Retrieval-Augmented Generation (RAG) with fine-tuning (FT) in the environmental domain, with the goal of enhancing the interpretation and generation of pollution control texts. The proposed hybrid system was evaluated on documents from local environmental agencies, using a combination of automated metrics (e.g., ROUGE, BLEU, and BERTScore) and preliminary expert assessment. The results indicate that the hybrid model achieved higher scores than other models on both automated metrics and expert evaluations. This suggests that it can effectively interpret and coherently generate technical guidance related to pollution control. These findings demonstrate the model's potential to support more reliable policy comprehension and implementation in environmental management, contributing to the intelligent interpretation of environmental policies and the advancement of data-driven environmental governance.

1. Introduction

With the accelerating pace of global industrialization and urbanization, environmental pollution has become increasingly severe, posing a critical challenge to sustainable development (Liu et al., 2020) . Recent scientific evidence further illustrates the growing complexity of contemporary pollution scenarios. For example, Zhang et al., (2025) revealed significant interactive toxicity effects between lipophilic and proteinophilic pollutants, highlighting the intricate chemical interactions and high-density technical knowledge involved in modern environmental pollution processes. In this context, pollution prevention and control strategies have emerged as central topics in policy formulation and scientific research across countries. Governments and international organizations have successively issued a series of regulations, industry standards, and technical guidelines aimed at enhancing pollutant emission control and improving environmental quality (Feng et al., 2024), despite the growing comprehensiveness of these policy frameworks, their practical implementation remains constrained by multiple factors—most notably, the complexity and limited applicability of pollution control technology guidelines (Lo et al., 2020).

Taking China as a representative example, the Ministry of Ecology and Environment has recently issued a substantial number of technical documents spanning industrial waste gas, wastewater treatment, and soil remediation, among others(Liu et al., 2020). These documents are typically dense with specialized terminology, logically structured, and hierarchically organized. Nevertheless, their practical promotion and field-level implementation frequently encounter common challenges, such as information overload, high cognitive barriers, and poor contextual adaptability (Liu et al., 2020; Liu et al., 2023). Many local environmental administrators, technical personnel in enterprises, and even members of the public find it difficult to accurately comprehend the intent of these policies. Consequently, the misalignment between technical measures and their practical implementation standards undermines the effectiveness of policy deployment and governance performance (Bartholomew et al., 2008).

Meanwhile, although pollution control technologies continue to evolve, and intelligent management and data-driven decision-making are emerging as prevailing trends (Bhupendra and Sangle, 2015),

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frontline environmental management still heavily relies on experiential judgment and manual knowledge transfer (Wang et al., 2023). The lack of structured processing and deep semantic understanding of policy-related texts has become a key bottleneck in advancing toward intelligent and precise environmental governance. The lack of standardized interpretation may hinder consistent implementation of policy documents and poses challenges to the development of systematic, evidence-based governance frameworks.

In recent years, artificial intelligence—particularly Large Language Models (LLMs)—has achieved remarkable progress in natural language understanding and generation tasks(Wang et al., 2025a; Tian et al., 2025). These models have been successfully applied in highly specialized domains such as healthcare (Haltaufderheide and Ranisch, 2024), law (Adhikary et al., 2024), education (Shen, 2024), demonstrating strong capabilities in knowledge modeling and semantic representation (Chen et al., 2025). However, several difficulties persist when applying LLMs to the pollution control domain. As prior studies have shown, legal and regulatory texts pose unique challenges for NLP systems due to their extremely long length, nested hierarchical clause structure, frequent cross-references and complex document layout (e.g., tables, emission limits, regulatory thresholds)(Katz et al., 2023). Standard pre-trained LLMs or generic fine-tuned models often underperform on such documents, especially in tasks requiring precise factual grounding and long-context reasoning (Khatri et al., 2025). Moreover, when documents combine unstructured text and dense tabular/structured data, even fact verification becomes challenging for NLP models (Chen et al., 2019).

To address these challenges, the integration of retrieval-augmented generation (RAG) and fine-tuning (FT) methods has been proposed in some studies. While hybrid LLM architectures are increasingly adopted in adjacent domains—primarily law and medicine, where tasks are often centered on short-form question answering or case-based reasoning—their application to environmental policy texts presents distinct and understudied challenges. In contrast to these settings, pollution control guidelines are characterized by unique features, including long and highly structured documents, interlinked regulatory references, and critical tabular data. These characteristics demand a tailored adaptation of the hybrid "RAG + FT" architecture that is specifically optimized for environmental semantic understanding. Specifically, our framework adopts a sequential RAG-to-generation pipeline, in which structure-aware retrieval from long regulatory documents provides mandatory evidence to a domain-specific fine-tuned generator, rather than treating retrieval and fine-tuning as loosely coupled components.

This work makes three key contributions. First, we develop and evaluate hybrid LLMs that combine RAG and FT, tailored to the technical and legal complexity of environmental regulatory texts. Second, we benchmark their performance against baseline and advanced models using both automated metrics and preliminary expert assessments, providing a more robust evaluation of domain comprehension. Third, we demonstrate the practical value of such systems for supporting policy interpretation, compliance, and data-driven environmental governance. Collectively, these contributions provide empirical evidence that hybrid LLMs can serve as a scalable and effective tool for intelligent environmental policy management.

2. Literature review

This section systematically reviews recent advances in the application of Large Language Models (LLMs) for domain-specific text processing, with a particular focus on their deployment in the environmental domain. Special attention is given to two key enabling techniques—Retrieval-Augmented Generation (RAG) and Domain-Specific Fine-Tuning—which are essential for understanding complex, structured texts. The review of existing achievements and research gaps lays a theoretical foundation for the integrated modeling framework proposed in this study.

2.1. Applications of large language models in the environmental field

In recent years, artificial intelligence has also been increasingly employed to support decision-making in environmental governance and policy-related economic mechanisms. For instance, Luo et al. (Luo et al., 2024) proposed a Deep Neural Network–based decision-making framework for carbon trading, demonstrating how data-driven models can assist in managing complex environmental economic systems under uncertainty. Such studies highlight the growing role of intelligent algorithms in environmental governance. However, the application of traditional machine learning or reinforcement learning methods remains limited in scalability and generalizability, particularly when dealing with the complex textual and knowledge structures inherent in pollution control technologies (Tu et al., 2024). With the emergence of large-scale pre-trained language models such as GPT-4, PaLM, and Qwen, LLMs have demonstrated unprecedented capabilities in language modeling across a variety of NLP tasks, including text comprehension, question answering, generation, and knowledge extraction (Banik et al., 2024). Owing to their generalized semantic understanding, LLMs exhibit strong transferability across global issues, making them particularly well-suited to multilingual and cross-regional challenges such as environmental pollution and climate change (Xia et al., 2023; Wang et al., 2025b). For example, Tu et al., (2024) proposed the use of LLMs for life-cycle inventory modeling, a similarly data-intensive task.

However, due to the broad and heterogeneous nature of their pre-training corpora—typically sourced from large-scale internet data—general-purpose LLMs often lack the ability to represent specialized knowledge effectively (Rawte et al., 2023). When tasked with interpreting highly structured and terminology-dense documents such as environmental policies or technical guidelines—which are often characterized by extreme length, hierarchical organization, and dense cross-references—LLMs frequently suffer from insufficient knowledge retrieval capacity, semantic deviation, and even hallucinations (Wu et al., 2024; Rawte et al., 2023). While several studies have begun to explore fine-tuning and knowledge-enhancement approaches in vertical domains like healthcare, law, and finance to improve domain adaptation (Castro et al., 2024), research in environmental governance remains fragmented and lacks a unified technical framework and evaluation methodology—particularly for high-stakes documents such as regulations and enforcement protocols (Nie and Liu, 2025).

2.2. Applications of retrieval-augmented generation (RAG) in specialized domains

Lewis et al., (2020) introduced the original RAG framework and proposed it as a means to enhance LLMs' ability to access and manipulate external knowledge. By integrating a document-level retrieval module into the generation process, RAG significantly improves factual accuracy and mitigates hallucination issues often encountered in open-domain tasks. RAG has since been successfully applied to a range of specialized domains, including medical question answering, legal document generation, and multi-hop reasoning (Wiratunga et al., 2024). For instance, Jeong et al., (2024) employed RAG to enhance the reasoning capabilities of medical LLMs by dynamically providing semantic context prior to generation, thus improving the factuality and relevance of model outputs.

Nonetheless, the application of RAG in domains characterized by highly structured and dense texts, such as pollution control technologies, presents several unresolved challenges: (1) how to construct high-quality, domain-specific knowledge bases with broad coverage (Barnett et al., 2024); (2) how to design retrieval and chunking strategies aligned with technical logic (Barnett et al., 2024); and (3) how to maintain semantic consistency and avoid redundancy or contradiction across multi-paragraph documents (Fan et al., 2024). These limitations constrain the full potential of RAG in intelligent environmental text processing.

2.3. Fine-tuning techniques and domain adaptation strategies

Pollution control technical documents typically suffer from two key challenges: data scarcity (Yang et al., 2025) and semantic complexity (Yu et al., 2024). These documents are often rich in specialized terminology, heavily context-dependent, and structurally intricate, which makes it difficult for general-purpose LLMs to be directly deployed without substantial risk of semantic drift, misinterpretation, or logical inconsistency (Hu et al., 2021).

In response, Parameter-Efficient Fine-Tuning (PEFT) techniques have emerged as a promising direction for domain adaptation. These methods enable effective transfer and knowledge injection without extensively modifying core model parameters, balancing generalization performance with computational efficiency. One representative approach is Low-Rank Adaptation (LoRA), which introduces low-rank decomposition modules to adjust select parameter layers—such as attention projections—thereby reducing training complexity and enhancing model adaptability under few-shot conditions(Hu et al., 2021). For instance, Feng et al., (2025) applied LoRA to fine-tune LLMs for provincial-level industrial carbon emissions forecasting, demonstrating its feasibility in environmental data modeling.

Given the complex syntactic structures and domain-specific terminology of environmental policy texts, researchers have also begun integrating prompt engineering (Zamfrescu-Pereira et al., 2023) and chain-of-thought (CoT) reasoning mechanisms (Wei et al., 2022) into fine-tuning pipelines to guide LLMs in recognizing contextual meanings and internal logical structures. While these strategies have shown promise in specialized tasks such as legal Q&A, regulatory generation, and evidence alignment, their application to pollution control texts—a uniquely interdisciplinary and structurally complex domain—remains in a nascent stage.

2.4. Integrating retrieval-augmented generation and domain-specific fine-tuning

Both RAG and PEFT-based fine-tuning address key limitations of LLMs in real-world applications: RAG enhances factual grounding through external knowledge retrieval (Fan et al., 2024), while fine-tuning techniques such as LoRA, Adapter Tuning, and Prefix Tuning enable cost-effective domain adaptation and terminology sensitivity (Hu et al., 2021).

Despite their respective advantages, these approaches often remain insufficient when applied in isolation to the multifaceted, cross-domain, and structurally complex tasks characteristic of pollution control texts. On one hand, such documents encompass heterogeneous information—including legal provisions, technical parameters, and environmental science terminology—with high information density and domain specificity. External retrieval may not be sufficient to ensure output accuracy or internal consistency in all cases(Jin and Dai, 2025). On the other hand, fine-tuning without robust contextual knowledge may lead to semantic drift or comprehension gaps (Wang et al., 2024). Therefore, constructing a hybrid architecture that synergizes both techniques is a critical direction for advancing intelligent environmental text processing.

Currently, research on the integration of RAG and fine-tuning remains in its early stages. For example, Lee et al., (2024) applied both RAG and fine-tuning strategies to explore LLM performance in architectural safety knowledge management, but limited their analysis to comparative benchmarking without designing a cohesive hybrid framework or assessing adaptability in structurally complex documents. However, within the context of environmental governance and pollution control, systematic investigations into how retrieval and fine-tuning can be jointly structured to address long, hierarchical, and regulation-intensive documents remain limited, with no widely established task definitions, modeling workflows, or evaluation protocols.

To address these gaps, this study proposes a unified modeling framework that integrates structure-aware retrieval with domain-specific fine-tuning. Specifically, a structured knowledge base of pollution control technologies is incorporated as an external retrievable resource, and LoRA-based fine-tuning is applied on a high-quality, small-scale professional corpus. This hybrid approach enhances the model's ability to accurately capture domain-specific terminology and comprehend complex contextual structures. The proposed strategy strengthens the relevance and factual alignment of generated content while mitigating semantic drift in multi-source information integration, offering high scalability and practical deployment potential.

3. Methodology

This study proposes an integrated framework that combines Retrieval-Augmented Generation (RAG) and domain-specific fine-tuning to process technical texts related to pollution control. The framework consists of four core components: data processing, model construction, hybrid inference, and performance evaluation. Given that all training and evaluation datasets in this study are in Chinese, Qwen3-8B (Alibaba, 2024) is selected as the base model. Beyond language compatibility, this choice reflects a deliberate trade-off between model capacity, computational efficiency, and practical deployability. The Qwen series has demonstrated strong and competitive performance across a wide range of Chinese and multilingual benchmarks, particularly in long-context understanding and reasoning tasks, making it well suited for regulatory and policy-oriented text processing. Compared with substantially larger open-source models such as Qwen3-72B, the 8B-parameter variant enables stable parameter-efficient fine-tuning and retrieval-augmented inference under realistic computational constraints, while maintaining competitive performance. In addition, relative to alternative Chinese LLM architectures such as ChatGLM, Qwen3 provides more flexible support for long-context inputs and PEFT-based fine-tuning frameworks, which is critical for handling the hierarchical structure and document length characteristic of pollution control guidelines.

3.1. Overall Research Framework

Fig. 1 illustrates the overall workflow of the proposed experimental framework. The process begins with the collection of pollution control technical guidelines and regulatory documents issued by the Ministry of Ecology and Environment of the People's Republic of China. These documents, which cover a wide range of industries, regions, and production processes, are cleaned and pre-processed to create two distinct datasets: one for the RAG model and another for the fine-tuning process.

In the second stage, a RAG model is constructed alongside a supporting knowledge base, using the same set of technical guidelines to generate question-answer (QA) pairs for fine-tuning the large language model. The knowledge base is structured via parent-child chunking techniques to ensure semantic coherence and retrieval granularity, while prompt engineering is employed to optimize QA pair extraction.

In the third stage, the outputs of the RAG model and the fine-tuned model are integrated into a composite reasoning system, which benefits from both dynamic knowledge retrieval and domain-specific language understanding.

Finally, evaluation is conducted using a set of pollution control guidelines publicly released by provincial and municipal environmental agencies across China. From these documents, a subset of QA pairs is extracted to form the test dataset. Model performance is assessed through a quantitative evaluation framework incorporating multiple metrics of accuracy and relevance. The study compares four systems: (1) a baseline general-purpose LLM, (2) the RAG-only model, (3) the fine-tuned-only model, and (4) the integrated RAG + fine-tuned model, to assess their respective effectiveness in handling complex environmental policy texts.

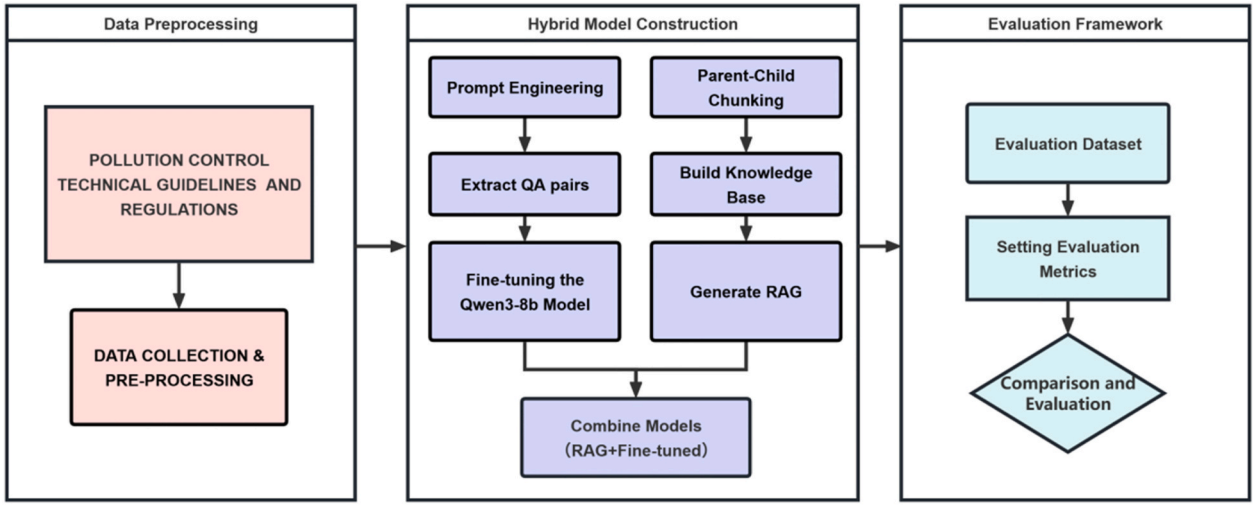


Fig. 1. Overall Research Framework. As shown in the diagram, the framework proceeds from data collection and preprocessing to hybrid model integration and final evaluation, incorporating both the RAG and fine-tuned components built on Qwen3-8B.

3.2. Data preprocessing

The quality of the dataset plays a decisive role in the performance of both the RAG model and the fine-tuned language model. Utilizing raw, unstructured text directly to construct a retrieval knowledge base can lead to semantic drift or information loss during upstream retrieval and downstream generation. Similarly, improperly formatted QA pairs can undermine the effectiveness of supervised fine-tuning and reduce generation accuracy.

To optimize model performance, this study employs distinct data processing strategies tailored to each model type. Both strategies operate on the same policy document corpus but apply two customized processing pipelines: a hierarchical clause-segmentation and chunking mechanism for the RAG system, and a structured QA pair construction method for fine-tuning. These procedures enhance data usability and adaptability while preserving essential domain semantics. To ensure regulatory alignment and eliminate ambiguities introduced by international or non-Chinese policy sources, a strict filtering protocol was implemented during data preprocessing. The retrieval corpus and fine-tuning datasets were restricted exclusively to authoritative pollution control regulations and technical guidelines issued by the Ministry of Ecology and Environment of the People's Republic of China and its provincial counterparts. This restriction guarantees that contextual grounding reflects locally applicable compliance obligations. Furthermore, when multiple versions or overlapping regulatory clauses were encountered, priority was assigned using a combination of regulatory authority level and issuance recency. This mechanism mitigates interpretive conflicts arising from heterogeneous or evolving regulatory sources. Combined with clause-level segmentation, the preprocessing strategies help preserve multi-level regulatory dependencies and support consistency in downstream retrieval and generation.

3.2.1. Construction of the RAG knowledge base

Due to the inherent limitations in the context window of large language models, the RAG architecture requires a high-quality knowledge base for effective performance. The structural organization of this knowledge base is directly tied to the precision of retrieval and the fluency of text generation.

In this study, a Parent-Child Chunking strategy (Carroll, 2010) combined with hybrid retrieval techniques (Hsu and Tzeng, 2025) is adopted instead of building a conventional Knowledge Graph (Xu et al., 2024), for the following key reasons: First, knowledge in the domain of pollution control technology primarily exists in unstructured textual form. Parent-child chunking preserves original semantic context better than structured graphs, thus improving the coherence and relevance of generated content (Zhang et al., 2024). Second, the construction of knowledge graphs involves complex entity recognition and graph maintenance, resulting in high development costs and deployment difficulties (Zhou et al., 2025). In contrast, parent-child chunking can be seamlessly integrated into existing RAG workflows with higher engineering feasibility. Finally, when paired with hybrid retrieval strategies, this method exhibits superior performance in semantic recall and generation accuracy, making it particularly suitable for question answering tasks centered on regulatory documents(Hsu and Tzeng, 2025).

As shown in Fig. 2, the knowledge base construction process begins with the development of a custom data disassembly and annotation pipeline based on Dify (Dify, 2023), an open-source framework available on GitHub. To adequately represent the complexity of Chinese environmental regulatory frameworks, we employed QwenMax (Alibaba, 2025), a Chinese LLM optimized for semantic understanding, to segment raw regulatory documents into hierarchical clause units following a chapter–article–subclause structure. During preprocessing, each clause was annotated with regulatory keywords and mandatory-action indicators to preserve structural dependencies and facilitate regulatory-strength–aware retrieval. In this study, the parent–child chunking strategy is implemented with explicitly defined and fixed hyperparameters to ensure reproducibility. Parent chunks are constructed as semantically self-contained document segments with a maximum length of 4000 tokens, designed to preserve complete regulatory or technical contexts. Each parent chunk is further divided into multiple child chunks with a maximum length of 500 tokens. These child chunks serve as the fundamental units for dense retrieval, with a 20 % overlap between adjacent child chunks to mitigate potential boundary information loss. This configuration enables fine-grained semantic matching while maintaining alignment with higher-level contextual information. During the retrieval phase, only child chunks are embedded and indexed in the vector database. Given a user query, candidate child chunks are first retrieved via dense vector similarity and subsequently reranked using a cross-encoder model. For response generation, the parent chunk corresponding to each retrieved child chunk is dynamically incorporated into the input context of the generator. Importantly, parent chunks are used exclusively for contextual grounding during generation and do not participate in retrieval or reranking. This approach enhances retrieval granularity while preserving contextual integrity, thereby significantly boosting the recall rate of the Retriever module and the coherence of the Generator module in content generation.

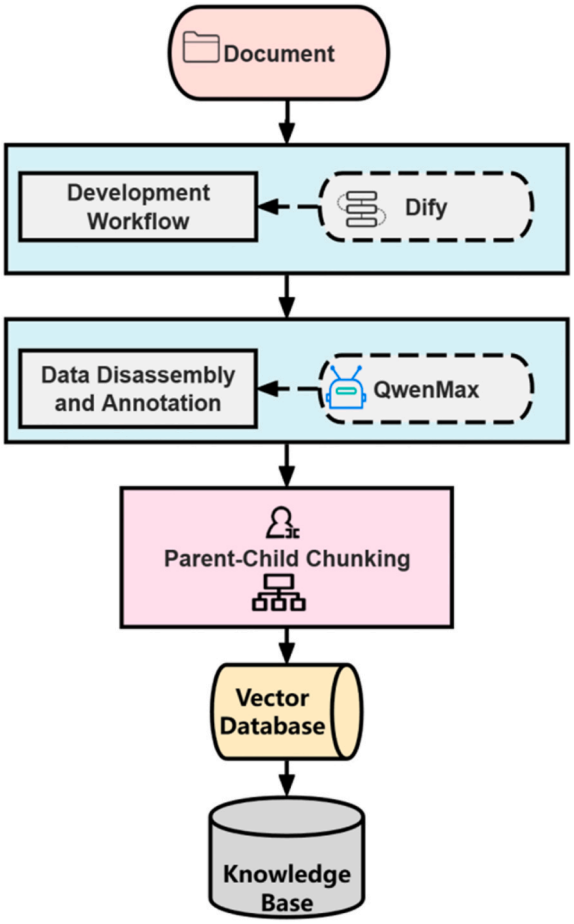


Fig. 2. Knowledge Base Construction Workflow
Note: The Dify-based text segmentation workflow has been uploaded to GitHub.

3.2.2. Fine-tuning QA dataset

For fine-tuning large language models on question-answering tasks, it is crucial to construct a structured and accurate QA-format dataset. Given the lack of publicly available QA resources specifically targeting Chinese pollution control technologies, this study builds a dedicated training corpus based on the technical standards and guidelines issued by the Ministry of Ecology and Environment.

To maximize data quality and reduce the risk of model hallucination, a Prompt Engineering (Zamfrescu-Pereira et al., 2023) approach was employed to generate QA pairs. The generation strategy integrates two advanced techniques: Chain-of-Thought (CoT)(Wei et al., 2022) reasoning and Self-Consistency (Brown et al., 2020). Chain-of-Thought prompts simulate human reasoning paths, guiding the model to generate step-by-step outputs, which significantly improves answer accuracy for logic-intensive or regulation-oriented questions. Self-Consistency performs multiple independent generations for the same input and then filters the results to retain only the most consistent outputs, enhancing the reliability and coherence of the training data. It should be noted that Chain-of-Thought reasoning is used only during QA data construction and supervision, and is not employed during model inference or evaluation.

Using the QwenMax model, prompt-based QA generation was applied to selected pollution control documents. After a combination of automated filtering and human validation for quality assurance, a final dataset of 3954 unique, high-quality QA pairs was compiled. Each QA item was crafted to correspond directly to individual policy clauses, ensuring strong alignment with the structure of source documents and high domain specificity (see Table 1).

Table 1
Examples from the fine-tuning QA dataset.

Question	Answer (Summary)	CoT (Summary)
How should hazardous wastes such as spent chromium-based catalysts and waste engine oil be safely disposed of to comply with legal requirements?	Hazardous wastes should be handled by certified entities in accordance with the Standard for Pollution Control on Hazardous Waste Storage (GB, 18597) and related laws to prevent environmental and legal risks.	identify risk → Emphasize certified units → Align with regulatory standards → Minimize legal and environmental risks
What treatment steps are required before discharging initial rainwater into municipal wastewater systems?	The water must undergo sedimentation and filtration to remove suspended solids and pollutants, ensuring it meets centralized treatment standards.	Identify water quality issue → Conduct sedimentation → Filter impurities → Reduce treatment plant burden

Note: The full dataset and Prompt Templates are provided in Supporting Information.

3.3. Model development

3.3.1. RAG model

Fig. 3 illustrates the architecture of the Retrieval-Augmented Generation (RAG) model developed in this study. To enable efficient interaction between the Qwen3-8B language model and the retrieval components, a hybrid retrieval pipeline based on the Dify framework was implemented. This pipeline integrates dense vector retrieval and full-text lexical retrieval, while prompt engineering is used to guide the generative behavior of the model. The previously constructed pollution control technical knowledge base serves as the primary source of contextual information.

During inference, user queries are first embedded into dense vectors using the BGE-M3 embedding model(Chen et al., 2024). These vectors are then compared against the knowledge base using cosine similarity to retrieve the most relevant content. Simultaneously, the system performs lexical retrieval using the BM25 algorithm (Robertson and Walker, 1994) to obtain supplementary documents based on keyword overlap. The results from both retrieval streams are combined and passed to a BGE-Reranker (Yang et al., 2025), which performs semantic reranking to improve the alignment between the retrieved documents and the user's query. Finally, the top-ranked documents are concatenated and, along with the original query, formatted via a prompt template before being passed to the Qwen3-8B inference model. This setup produces well-structured, semantically accurate answers tailored to the technical and regulatory context of pollution control.

To prioritize and weight retrieved sources, the pipeline applies a two-stage relevance scoring mechanism. Dense embedding similarity is first computed to identify semantically related candidate clauses. Next, a cross-encoder reranking model evaluates contextual relevance at a finer granularity, producing final relevance scores for all retrieved passages. The Top-K clauses incorporated into the prompt are selected according to these reranked scores rather than lexical frequency or keyword overlap. Through this ranking-based selection, regulatory clauses with higher semantic specificity and stronger contextual alignment are preferentially included during generation, allowing the RAG + FT framework to integrate relevant regulatory fragments coherently even when multiple candidate clauses coexist.

To enhance transparency and traceability during inference, retrieved clauses included in the prompt remain visible as structured context, enabling a direct correspondence between model outputs and their regulatory sources. Because QA responses are constructed using explicitly retrieved regulatory text rather than latent knowledge alone, users can examine which retrieved clauses support each answer. This grounded generation process improves transparency around answer

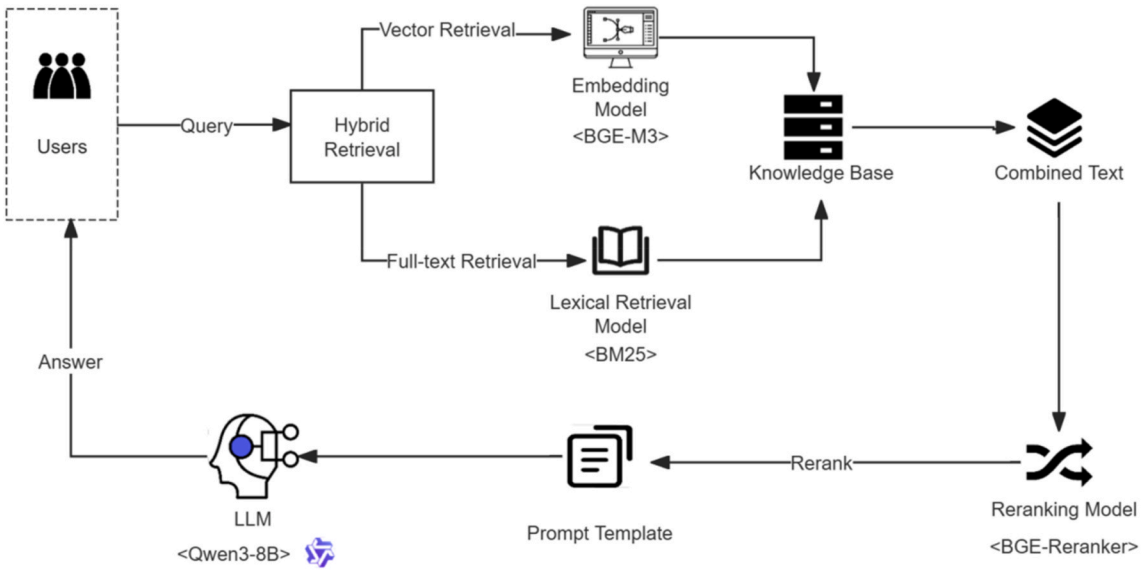


Fig. 3. RAG Architecture Diagram. Depicts hybrid retrieval with vector + lexical search, semantic reranking, and final generation using Qwen3-8B LLM.

provenance and regulatory grounding and supports user verification of compliance-related reasoning.

3.3.2. *Fine-tuning the model*

This study adopts the Llama-Factory (Zheng et al., 2024) framework to implement efficient fine-tuning, drawing methodological reference from the work of Zheng et al., (2024). To maintain experimental consistency, Qwen3-8B was selected as the base model for fine-tuning. The process employs the Parameter-Efficient Fine-Tuning (PEFT) approach, specifically Low-Rank Adaptation (LoRA), which freezes the majority of the model's parameters while selectively updating a small subset. This is achieved by decomposing weight matrices into lower-rank formats.

LoRA significantly reduces computational and memory overhead and is particularly effective in settings with limited training data. Given the complexity and scale of the Qwen3 architecture, this method enables the model to be efficiently adapted to the pollution control domain without necessitating extensive retraining or compromising performance. In addition, model quantization was intentionally omitted to preserve precision during fine-tuning.

Training was conducted using the AdamW optimizer, with hyperparameters set as $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 1e\text{-}8$. A cosine learning rate scheduler was employed to ensure a smooth optimization process, with the initial learning rate set to 5e-5. To accommodate hardware limitations, gradient accumulation was configured at 8 steps, resulting in an effective total batch size of 16, based on a per-device batch size of 2.

After multiple rounds of experimentation, the optimal performance was achieved after 3 epochs. As shown in Fig. 4, the model's loss function consistently decreased over time, indicating a stable training process. The figure presents both the original loss curve and its smoothed trend line, clearly demonstrating convergence. The initial loss of approximately 1.45 steadily declined to below 0.8 after around 750 training steps, confirming effective optimization without overfitting.

Summary of Fine-Tuning Hyperparameters:

Initial Learning Rate: 5e-5.

Optimizer: AdamW.

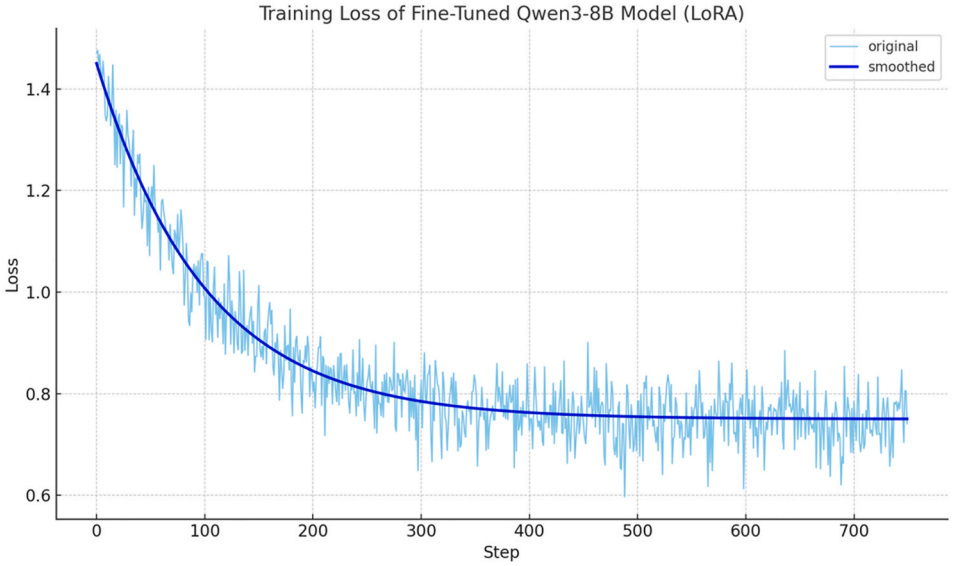


Fig. 4. Training Loss by Epoch. Displays the original and smoothed training loss curves for LoRA-based fine-tuning on the Qwen3-8B model.
Note: The fine-tuned model has been uploaded to Hugging Face. Refer to Supporting Information for further details.

Learning Rate Scheduler: Cosine.
Number of Epochs: 3.
Gradient Accumulation Steps: 8.
Per-Device Training Batch Size: 2.
Per-Device Evaluation Batch Size: 8.
Random Seed: 42.

The hyperparameters listed above were selected via a lightweight, manually guided tuning process rather than an extensive grid or Bayesian search. We varied the learning rate, number of epochs, and effective batch size on a held-out validation split and chose the configuration that provided stable training and good validation performance, and then applied this configuration consistently to both the FT and RAG + FT models.

3.4. Hybrid model design

To further enhance the performance of text generation tasks in the context of pollution control technologies, a hybrid architecture was developed by integrating Retrieval-Augmented Generation (RAG) with a fine-tuned large language model. This design aims to combine the strengths of the RAG module in external knowledge retrieval with the fine-tuned model's capability for domain-specific language understanding, thereby improving both semantic accuracy and linguistic quality.

The hybrid model operates in a two-stage process:

Stage 1: The input question is processed by the RAG module, which retrieves the most relevant textual fragments from a knowledge base using vector-based retrieval and reranking mechanisms.
Stage 2: The retrieved content, along with the original question, is passed to the LoRA-fine-tuned Qwen3-8B model. This model has been adapted using domain-specific technical corpora in pollution control, endowing it with enhanced domain modeling capabilities.

Final Output: The fine-tuned model integrates contextual input with specialized knowledge to generate answers with higher accuracy and domain relevance.

This hybrid approach ensures both the completeness of factual knowledge and the precision of language generation. Common issues observed in single-strategy models, such as semantic drift, logical discontinuity, and terminology misapplication, are effectively mitigated. Experimental results indicate that the hybrid model significantly outperforms standalone RAG or fine-tuned models across lexical overlap and semantic evaluation metrics.

3.5. Evaluation framework

3.5.1. Evaluation dataset

The evaluation dataset was constructed to assess the applicability of the proposed models to pollution control scenarios. Source data were collected from pollution control regulations independently published by provincial and municipal environmental protection authorities in China. From this corpus, 100 high-quality entries were manually selected as representative test samples. These samples were excluded from the model development process to ensure the validity of the evaluation. To ensure robust and unbiased evaluation, a k-fold cross-validation strategy (k = 5) was adopted. The full evaluation dataset was randomly partitioned into five non-overlapping subsets. In each fold, four subsets were used for model inference and one subset for testing. This process was repeated five times, and all reported results represent the mean ± standard deviation across folds.

Example:

Question: “In VOC emissions control, how are total VOCs and non-methane hydrocarbons categorized as control items?"

Answer: “In VOC emissions control, total VOCs (TVOC) and non-methane hydrocarbons (NMHC) are used as representative indicators for overall VOC emissions. TVOC reflects the comprehensive emission level, while NMHC is used for specific compound regulation. These metrics are jointly applied to monitor and evaluate VOC emissions compliance."

Note: The full evaluation dataset is available in Supporting Information.

3.5.2. Multi-dimensional evaluation metrics

To comprehensively assess model performance on pollution control text generation tasks, a set of multi-dimensional quantitative metrics was applied. These metrics are grouped into three major categories:

Lexical Overlap Metrics: Including ROUGE-1, ROUGE-2, ROUGE-L (Lin, 2004) as well as BLEU (Papineni et al., 2002) are employed to measure n-gram overlap and longest common subsequence similarity between generated and reference texts. In addition, token-level overlap Precision, Recall, and F1-score (Wang et al., 2022) are computed to measure, respectively, the proportion, coverage, and harmonic mean of overlapping tokens between the generated outputs and the reference answers. Together, these metrics provide a comprehensive characterization of surface-level lexical fidelity.

Semantic Similarity Metrics: Cosine similarity between sentence embeddings of generated and reference texts was computed using the large-scale independently pretrained E5-large-v2 embedding model (Wang et al., 2022). This evaluation embedding model is architecturally and pre-training independent from the retrieval embedding model (BGE-M3) used in the RAG pipeline, thereby avoiding potential evaluation bias and ensuring the objectivity and robustness of semantic similarity assessment.

BERTScore Metrics: Derived from vector representations of a pre-trained BERT model (Devlin et al., 2019), this set includes BERT-based Precision, Recall, and F1 to assess the preservation, coverage, and overall integrity of semantic content.

All metrics were computed on a standardized test set across five model configurations: GPT-4o, Qwen3-8B (base), RAG, Fine-Tuned, and RAG + Fine-Tuned. From an architectural perspective, the FT-only and RAG-only variants can be viewed as ablation-style baselines of the full RAG + FT model, isolating the respective contributions of domain-specific fine-tuning and retrieval augmentation.

3.5.3. Normalization

To ensure comparability across metrics with different ranges and units, a min–max normalization method was applied to standardize raw scores. The normalization was performed using the following formula:

$$s_i' = (s_i - s_{min}) / (s_{max} - s_{min})$$

s_i is the raw score for a specific model and metric, s_{min} and s_{max} are the minimum and maximum scores across all models for that metric, $s_i' \in [0, 1]$ represents the normalized score, reflecting the relative performance of the model in that dimension.

In this study, normalized values are used exclusively for the radar-chart visualization (Fig. 8), to place heterogeneous metrics on a common [0, 1] scale for intuitive comparison. All numerical results reported in the tables and text (means, standard deviations, and relative improvements) are computed from the unnormalized metric values, and no additional scaling or weighting is applied in the statistical analysis.

3.5.4. Expert evaluation of model outputs

To address the limitations of automatic evaluation metrics (e.g., ROUGE, BLEU, and BERTScore) in capturing domain-specific understanding and application capabilities, and to further verify the factual accuracy and practical utility of model outputs, this study designed and implemented an expert-based evaluation component. This evaluation aimed to provide a more in-depth and reliable assessment of model performance from the perspectives of domain knowledge and real-world applicability.

(1) Expert Panel and Background

A panel of five domain experts was assembled for this study. The panel included two university professors with extensive experience in environmental policy and regulatory research, two engineers with professional expertise in pollution control technology and management, and one government-affiliated advisor experienced in interpreting pollution control standards. The experts' collective backgrounds cover environmental engineering, regulatory policy, and corporate compliance management, ensuring a comprehensive and authoritative evaluation process. To maintain objectivity, all experts rated the model outputs independently and anonymously, and they were blinded to the source model of each response during the evaluation process.

(2) Evaluation Dataset and Model Scope

The expert evaluation was conducted on a dataset of 20 question–answer (QA) pairs manually selected from the test set, covering six different industrial sectors. These QA pairs represent common regulatory clauses, management requirements, and compliance standards found in pollution control guidelines. For each question, experts evaluated the responses generated by four types of models: (1) the base large language model (Qwen3-8B and GPT-4o), (2) the domain-specific fine-tuned model (FT), (3) the retrieval-augmented generation model (RAG), and (4) the integrated RAG + FT hybrid model.

(3) Evaluation Criteria and Scoring Standards

The expert evaluation questionnaire was designed based on domain knowledge requirements and compliance review practices in environmental management. Four evaluation dimensions were established, each rated on a five-point Likert scale (1 = poor, 5 = excellent), as follows:

Correctness: Whether the response accurately reflects current regulatory standards and requirements, and whether any factual errors are present.

Completeness: Whether the response covers all key information and essential points related to the question.

Compliance: Whether the response demonstrates sufficient technical depth and domain expertise, and whether it aligns with practical management and regulatory requirements.

Clarity: Whether the response is clearly expressed, logically coherent, and easy to understand and apply in practice.

To ensure the reliability of the expert-based evaluation, inter-rater agreement was explicitly examined across all evaluation dimensions. Specifically, Fleiss' kappa (Fleiss and Jacob, 1973) was computed separately for each dimension by pooling all evaluated samples across models. This analysis assesses whether experts apply stable and consistent judgment criteria when evaluating different model outputs, independent of model identity. In addition, to ensure comparability between expert-based scoring and automated evaluation metrics, all evaluation dimensions were normalized to a unified five-point scale, and results were reported per dimension rather than collapsed into a single weighted score. This design avoids subjective weighting and allows performance differences to be interpreted transparently across Correctness, Completeness, Compliance, and Clarity.

4. Results

4.1. Real-world task simulation and output comparison

Table 2 presents a comparative summary of responses generated by five large language models to the question: 'Which components of used lead-acid batteries need to be separately recycled and treated to prevent pollution during the dismantling process?' The reference answer, derived from authoritative policy documents, is grounded in environmental regulations and highlights the necessity of recovering plastic casings, lead plates, lead-containing plastics, and waste acid. It explicitly mandates that waste acid must not be discharged into the environment or smelted together with battery shells.

Table 2
Example of outputs of each model (content translated and simplified from the Chinese text).

Question	Which components of used lead-acid batteries need to be separately recycled and treated to prevent pollution during the dismantling process?
Reference Answer	Used lead-acid batteries must be dismantled to separately recover and treat plastics, lead plates, lead-containing materials, and waste acid. Particularly, waste acid must be collected and treated; it is strictly prohibited to discharge it into the environment or to smelt batteries containing acid.
GPT-4	Identifies treatment for electrolyte, lead paste, plastic casing, separators, and metal connectors, noting specific hazards and treatment techniques. The content is thorough but leans towards technical details based on international standards.
Qwen3-8B	Covers lead plates, electrolytes, plastic casings, terminals, and wastewater. Emphasizes principles of classification, sealing, neutralization, and recovery.
RAG	Highlights plastics, lead plates, lead-containing materials, and waste acid for separate treatment, stressing the risks of combined processing and advocating corresponding detoxification technologies.
Fine-Tuned	Points out the need to treat plastics, electrolytes, positive and negative plates, and separators. Stresses compliance with environmental standards to prevent secondary pollution.
RAG + FT	Identifies plastics, lead plates, lead-containing materials, and waste acid as key pollutant sources. Emphasizes proper collection, neutralization, and categorized smelting techniques. The response is professionally articulated and technically complete.

Note: The full dataset is provided in Supporting Information.

GPT-4 offers a structurally complete response and emphasizes the recovery of various detailed components along with associated risks and treatment techniques. However, some references appear to align more with international standards and may not fully conform to Chinese regulatory frameworks. The Qwen3-8B model provides a broad yet structured output, underscoring classification and environmental control principles, demonstrating moderate domain alignment. The RAG model, leveraging retrieval capabilities, precisely identifies key pollutant sources but tends to deliver concise content with some process-level treatments remaining underdeveloped. The Fine-Tuned model reflects an understanding of technical terms and environmental expression patterns, though its treatment descriptions are comparatively general. The RAG + Fine-Tuned model combines the contextual accuracy of retrieval with the linguistic precision of domain fine-tuning. Its output is the most comprehensive, structurally coherent, and compliant, demonstrating both an accurate grasp of the problem and professional articulation.

4.2. Performance gains in regulatory QA accuracy

Table 3 presents a comprehensive comparison of model performance under the refined evaluation protocol, covering n-gram overlap, token-level matching, and semantic similarity metrics. As shown, the RAG + Fine-Tuned (RAG + FT) model consistently achieves the strongest overall performance, attaining the highest or near-highest scores across most evaluation indicators.

Compared with the base Qwen3-8B model, RAG + FT delivers an average relative improvement of 63.8 % across classical generation and overlap metrics, including ROUGE-1, ROUGE-2, ROUGE-L, BLEU, and token-level Precision, Recall, and F1. These gains indicate substantial improvements in lexical alignment, answer completeness, and coverage of relevant regulatory content, highlighting the effectiveness of integrating retrieval augmentation with domain-specific fine-tuning.

At the semantic level, the hybrid model also demonstrates clear

Table 3
Performance comparison of models in pollution prevention tasks.

Type	Metric	GPT-4o	Qwen3-8B	RAG	Fine-tuned	RAG + FT
N-grams	ROUGE-1	0.1818 ± 0.0421	0.1920 ± 0.0521	0.2198 ± 0.0227	0.2857 ± 0.0385	0.3353 ± 0.0481
	ROUGE-2	0.0823 ± 0.0221	0.1172 ± 0.0418	0.1229 ± 0.0510	0.1738 ± 0.0628	0.2175 ± 0.0587
	ROUGE-L	0.1698 ± 0.0377	0.1844 ± 0.0513	0.2105 ± 0.0413	0.2871 ± 0.0277	0.2997 ± 0.0468
	BLEU	0.0017 ± 0.0007	0.0027 ± 0.0010	0.0026 ± 0.0011	0.0031 ± 0.0005	0.0034 ± 0.0008
Token-level overlap	Precision	0.0259 ± 0.0065	0.0446 ± 0.0174	0.0387 ± 0.0097	0.0542 ± 0.0121	0.0690 ± 0.0203
	Recall	0.1313 ± 0.0523	0.1436 ± 0.0423	0.1563 ± 0.0574	0.2218 ± 0.0509	0.2436 ± 0.0538
	F1	0.0383 ± 0.0076	0.0655 ± 0.0136	0.0622 ± 0.0101	0.0725 ± 0.0119	0.1139 ± 0.0141
Semantic	Cosine	0.8648 ± 0.0513	0.8628 ± 0.0478	0.9117 ± 0.0349	0.8913 ± 0.0415	0.9196 ± 0.0528
BERT	Precision	0.6351 ± 0.0414	0.6804 ± 0.0458	0.7438 ± 0.0521	0.7128 ± 0.0534	0.7915 ± 0.0819
	Recall	0.7294 ± 0.0246	0.7366 ± 0.0193	0.7628 ± 0.0342	0.7486 ± 0.0284	0.7930 ± 0.0597
	F1	0.6780 ± 0.0261	0.7060 ± 0.0272	0.7526 ± 0.0421	0.7292 ± 0.0302	0.7911 ± 0.0705

advantages. Specifically, RAG + FT achieves an average relative improvement of 10.5 % across semantic similarity metrics, including BERTScore Precision, Recall, F1, and Cosine Similarity, reflecting enhanced semantic fidelity and contextual consistency with reference answers.

Although the standard deviations of certain semantic precision-related metrics are moderately higher, this variability is attributable to the model's increased sensitivity to domain-specific regulatory expressions rather than instability. In contrast, GPT-4o exhibits relatively stable performance with lower variance across metrics; however, its consistently lower scores suggest limited adaptability to specialized regulatory QA scenarios compared with retrieval-augmented and fine-tuned models. These observations are specific to the Chinese pollution control QA setting examined in this study and should not be interpreted as a general statement about GPT-4o's performance in other domains.

4.3. Lexical fidelity and text structuring improvements

Fig. 5 illustrates model performance on lexical overlap and token-level matching metrics, including ROUGE-1, ROUGE-2, ROUGE-L, and token-level Precision, Recall, and F1, reported as mean values with standard deviations across K-fold cross-validation. Overall, the RAG + Fine-Tuned model demonstrates the strongest lexical fidelity, achieving the highest or near-highest scores across all ROUGE variants. In particular, its ROUGE-1 and ROUGE-L scores fall in the range of approximately 0.30–0.34, while ROUGE-2 consistently exceeds 0.21, indicating improved preservation of both unigram coverage and longer-span phrase structures. The Fine-Tuned model also shows substantial improvements over GPT-4o and the base Qwen3-8B model, with ROUGE scores generally ranging from 0.17 to 0.29, highlighting the effectiveness of domain-specific fine-tuning in enhancing lexical alignment with regulatory guidelines. In contrast, the RAG-only model yields moderate gains, achieving ROUGE scores in the 0.12–0.22 range, but remains less effective on token-level F1, reflecting its reliance on retrieved fragments without fully optimized generation. Although GPT-4o and Qwen3-8B exhibit relatively stable performance with smaller variances, their absolute scores are consistently lower—particularly on ROUGE-2, where scores remain below 0.12—underscoring the limitations of general-purpose models in capturing domain-specific lexical patterns. The error bars further indicate that the observed advantages of the hybrid RAG + Fine-Tuned model are consistent across folds rather than driven by isolated samples.

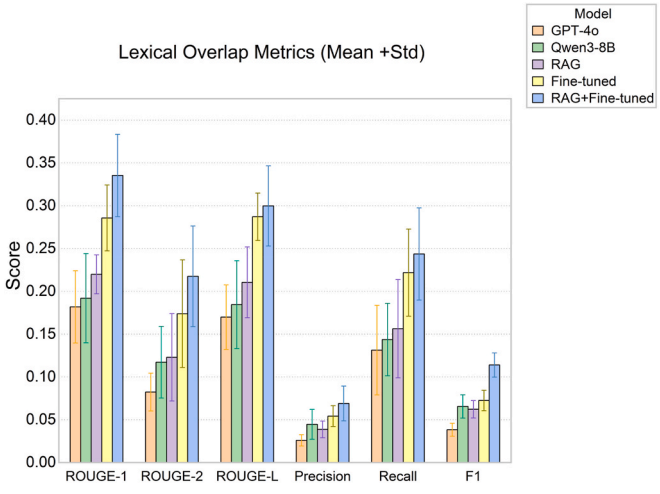


Fig. 5. Cross-validated Lexical Overlap Metric Comparison Across Models (Mean + Std)
Note: BLEU scores are omitted from the main figure due to their substantially smaller magnitude under the current evaluation setting and are reported separately in Supporting Information for completeness.

Beyond n-gram overlap, a qualitative inspection of sample outputs shows that the RAG + FT model more faithfully preserves the internal discourse structure of regulatory clauses—for example, the "scope–condition–requirement" ordering and the enumeration of multi-step procedures—whereas non-retrieval baselines more frequently merge, reorder, or partially omit such elements despite fluent generation.

4.4. Semantic consistency in guideline interpretation

Fig. 6 reports the semantic similarity between generated answers and reference texts measured by cosine similarity of contextualized embeddings, with results shown as mean values and standard deviations across K-fold cross-validation. Across all models, cosine similarity scores remain relatively high, indicating that most systems capture the overall semantic intent of regulatory guideline interpretations. Among them, the RAG + Fine-Tuned model achieves the highest average cosine

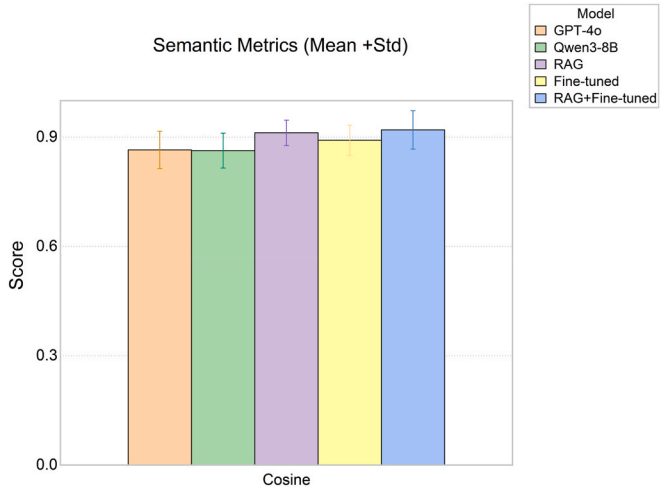


Fig. 6. Cross-validated semantic metrics comparison (mean + std).

similarity, with values exceeding 0.91, suggesting closer alignment in the global semantic space. The Fine-Tuned and RAG models follow closely, while GPT-4o and Qwen3-8B exhibit slightly lower scores. These results indicate that retrieval augmentation and domain-specific fine-tuning primarily improve global semantic consistency, ensuring that generated responses remain directionally aligned with the intended regulatory meaning, even when surface realizations differ.

4.5. Domain-specific understanding evaluated by BERTScore

Fig. 7 presents model performance measured by BERTScore Precision, Recall, and F1, which assess semantic similarity at the token level by evaluating the alignment and coverage of contextualized representations. As shown in the figure, the RAG + Fine-Tuned model achieves the highest mean BERTScore values across all three sub-metrics, with scores clustering around 0.79, indicating more comprehensive semantic coverage of reference regulatory texts. In comparison, the RAG and Fine-Tuned models obtain BERTScore F1 values in the range of approximately 0.73–0.75, suggesting that while much of the semantic content is preserved, certain fine-grained regulatory details remain partially uncovered. GPT-4o and Qwen3-8B exhibit lower BERTScore values, particularly in Precision and F1 (generally below 0.70), reflecting a greater degree of semantic omission or mismatch at the token level. These results highlight that improvements observed in cosine similarity do not necessarily translate into full semantic coverage, and that the hybrid approach is more effective in capturing detailed, domain-specific regulatory content.

In this configuration, domain-specific fine-tuning mainly improves semantic alignment by adapting the backbone representations and generation patterns to pollution-control terminology and clause structures, while retrieval augmentation in the RAG + FT model further enhances BERTScore Recall and F1 by conditioning answer generation on clause-level regulatory segments, thereby reducing omissions and paraphrasing drift.

4.6. Comprehensive multi-metric scoring and visualization

To obtain an integrated view of model performance across heterogeneous indicators, this study aggregates the eleven automatic metrics reported in Table 3 into a unified multi-metric visualization. For each metric (ROUGE-1/2/L, BLEU, token-level Precision, Recall and F1, cosine similarity, and BERTScore Precision, Recall and F1), the mean scores of the five models are first normalized to the [0,1][0,1][0,1] range using min–max scaling across models. In this representation, a value of 1.0 corresponds to the best-performing model on that metric and 0.0 to the worst, allowing direct comparison of relative strengths under a common scale.

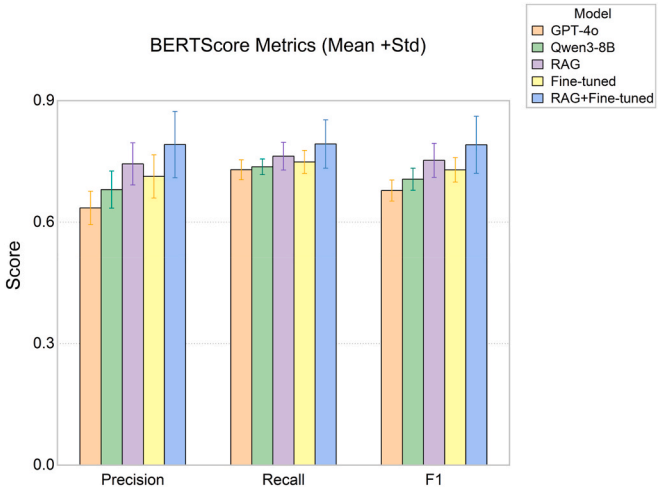


Fig. 7. Cross-validated BERTScore metrics analysis (mean + Std).

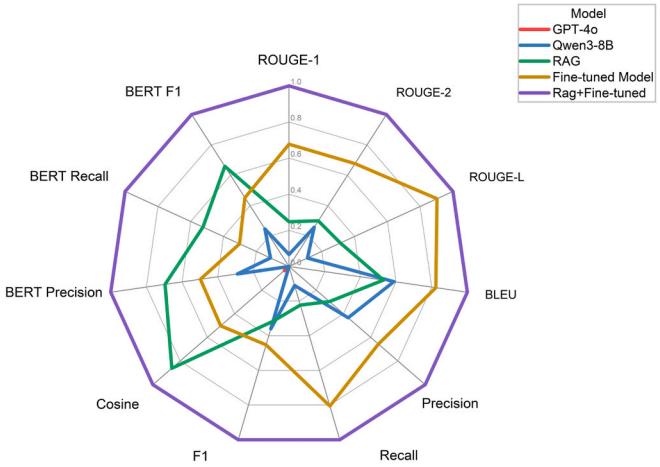


Fig. 8. Normalized radar chart of model performance across evaluation dimensions.

Fig. 8 displays the resulting normalized scores in a radar chart. The RAG + Fine-tuned model encloses the largest area and remains closest to the outer contour on nearly all axes, confirming its consistently strong performance across both lexical-overlap and semantic-similarity dimensions. In particular, its advantages are most pronounced on ROUGE-2, ROUGE-L, token-level F1, and BERTScore F1, indicating that the hybrid architecture is especially effective at preserving multi-token regulatory expressions, capturing long-range contextual dependencies, and aligning generated answers with the semantic content of reference guidelines.

The Fine-tuned model forms the second-largest polygon, reflecting clear gains from domain-specific supervision alone, especially on lexical metrics. The RAG-only model shows competitive performance on semantic-oriented indicators such as cosine similarity and BERTScore Recall, but its overall radar shape is less balanced, suggesting that retrieval without fine-tuning is more prone to omissions or over-reliance on retrieved fragments. By contrast, the baseline models GPT-4o and Qwen3-8B occupy the inner region of the radar chart on most axes, indicating weaker lexical fidelity and semantic coverage in the specialized setting of pollution-control QA, despite their strong general-purpose language capabilities.

Overall, the radar chart provides an intuitive, normalized comparison of the five models and visually highlights the comprehensive advantages of the RAG + Fine-tuned configuration across the full set of automatic evaluation metrics. To provide a more fine-grained view of model behavior and to reduce potential visual bias introduced by the radar representation, we additionally report a heatmap of the raw mean scores for all eleven automatic metrics across the five models in the Supporting Information (Fig. S1). This complementary visualization directly reflects the values in Table 3 and confirms the advantage of the RAG + FT model across both lexical and semantic metrics.

4.7. Expert-based evaluation of model performance

The expert evaluation results are presented in Table 4 Among all models, the RAG + FT hybrid model achieved the highest average score (4.08), demonstrating superior domain understanding and balanced performance across all four evaluation dimensions. In particular, it achieved the best performance in Compliance (4.1 ± 0.2) and Clarity (4.1 ± 0.2), highlighting the advantage of combining retrieval-augmented generation with domain-specific fine-tuning.

The RAG model (3.95) and FT model (3.93) followed closely, indicating that both retrieval augmentation and fine-tuning independently

Table 4
Expert evaluation scores (mean ± SD) of five models across four performance dimensions.

Model	Correctness	Completeness	Compliance	Clarity	Average Score
GPT-4o	3.7 ± 0.4	4.1 ± 0.3	3.1 ± 0.6	4.4 ± 0.3	3.83
Qwen3-8B	4.0 ± 0.3	3.7 ± 0.4	3.7 ± 0.4	4.0 ± 0.3	3.85
FT Model	4.0 ± 0.2	3.8 ± 0.4	3.9 ± 0.4	4.0 ± 0.2	3.93
RAG Model	4.2 ± 0.5	3.7 ± 0.3	4.0 ± 0.3	3.9 ± 0.5	3.95
RAG + FT Model	4.2 ± 0.3	3.9 ± 0.2	4.1 ± 0.2	4.1 ± 0.2	4.08

contribute to improved domain knowledge representation and answer quality. Notably, the retrieval-based approach slightly outperformed fine-tuning alone, reflecting its strength in contextual relevance and regulatory alignment.

Among the baseline models, Qwen3-8B (3.85) marginally outperformed GPT-4o (3.83), likely due to its optimization for Chinese-language regulatory texts. However, GPT-4o exhibited the highest Clarity score (4.4 ± 0.3) and strong Completeness (4.1 ± 0.3), which may be attributed to its substantially larger parameter scale and broader pre-training corpus compared with Qwen3-8B. This advantage enables GPT-4o to generate more fluent, coherent, and content-rich responses. Nevertheless, its Compliance score (3.1 ± 0.6) remained significantly lower. A qualitative inspection of expert comments reveals that the lower Compliance scores of GPT-4o are mainly attributable to two recurring error patterns. First, the model occasionally produces regulatory statements that are not explicitly present in the referenced Chinese pollution control documents, despite being linguistically plausible. Second, GPT-4o sometimes relies on generalized environmental management concepts, regulatory practices from non-Chinese jurisdictions, or outdated versions of regulatory provisions. These issues highlight the limitations of general-purpose LLMs in compliance-critical regulatory interpretation without domain-specific grounding.

To assess the reliability of expert judgments underlying these results, inter-rater agreement was quantified using Fleiss' kappa. Substantial agreement was observed across all evaluation dimensions, including Correctness ($\kappa = 0.71$), Completeness ($\kappa = 0.65$), Compliance ($\kappa = 0.61$), and Clarity ($\kappa = 0.76$). These results confirm that the observed performance differences are supported by consistent expert assessments rather than individual subjective bias.

Overall, these results confirm that integrating retrieval-augmented generation with domain-specific fine-tuning yields the most robust and domain-aligned performance, surpassing both baseline LLMs and single-component approaches. This suggests strong potential for deploying the RAG + FT framework in real-world applications for interpreting and implementing pollution control policies.

5. Discussion

5.1. Major findings and significance

This study introduces a hybrid large language model (LLM) framework that integrates Retrieval-Augmented Generation (RAG) and domain-specific fine-tuning (FT) to improve the interpretation and application of pollution control guidelines. The results demonstrate that the proposed RAG + FT approach consistently outperforms baseline and single-strategy models across multiple dimensions, including Correctness, Completeness, Compliance, and Clarity.

Expert evaluation results (Table 4) indicate that the RAG + FT model achieved the highest average score of 4.08. This represents a clear improvement over the baseline models GPT-4o (average: 3.83) and Qwen3-8B (average: 3.85). These findings suggest that combining domain knowledge adaptation with dynamic knowledge retrieval significantly enhances the model's capacity for deep contextual understanding and factual reasoning in complex regulatory environments.

In addition, this study highlights the complementary roles of automated metrics (e.g., ROUGE, BLEU, BERTScore) and expert-based evaluation. Automated metrics capture surface-level similarity and semantic consistency, whereas expert review provides deeper insights into domain correctness, regulatory compliance, and practical usability. By leveraging both approaches, the evaluation framework achieves a more comprehensive and scientifically robust assessment of model performance, offering methodological guidance for future research on LLM evaluation in policy and technical text interpretation.

5.2. Analysis of model performance differences

The performance differences observed across models reflect fundamental distinctions in their architecture design and knowledge adaptation strategies. The baseline model GPT-4o achieved the highest Clarity (4.4 ± 0.3) and strong Completeness (4.1 ± 0.3), likely due to its significantly larger parameter scale and extensive pretraining on broad, general-domain corpora. This endows GPT-4o with superior fluency and coverage in text generation. However, its relatively low Compliance score (3.1 ± 0.6) indicates limitations in deep domain reasoning and specialized knowledge application, constraining its effectiveness in tasks such as regulatory interpretation and technical detail understanding.

The Qwen3-8B model performed better in semantic modeling and knowledge alignment (average: 3.85), particularly for understanding and generating Chinese regulatory texts. Nevertheless, its lack of a dynamic knowledge retrieval mechanism restricts its ability to reason across complex, multi-document legal provisions and maintain contextual coherence in responses. Although Qwen3-8B demonstrates relatively strong semantic alignment and structured output generation, its ability to model domain-specific terminology remains limited. This is largely due to the context-dependent nature of environmental regulatory language, in which key technical terms participate in hierarchical and constraint-based relationships across multiple clauses and documents. Without explicit domain adaptation or regulation-aware retrieval, Qwen3-8B tends to interpret such terms using generalized semantic priors learned during pretraining, which can lead to subtle loss of regulatory precision or incomplete mapping between technical entities and compliance obligations. This limitation helps explain its performance gap relative to the RAG + FT hybrid architecture, whose retrieval and fine-tuning components jointly enhance grounding in domain-specific terminology and clause-level semantic dependencies.

Both single-strategy approaches demonstrated specific strengths. The FT model, by aligning the model's knowledge structures and semantic patterns with domain-specific data, improved Compliance (3.9 ± 0.4) and Correctness (4.0 ± 0.2). The RAG model, by dynamically incorporating authoritative external knowledge, mitigated semantic drift and improved Correctness (4.2 ± 0.5) and Compliance (4.0 ± 0.3). Ultimately, the RAG + FT model combines these advantages, ensuring factual accuracy and contextual relevance while enhancing the model's capacity to parse complex regulatory language and perform deeper technical reasoning.

These results underscore a critical insight: relying solely on large-scale pretraining or a single adaptation strategy is insufficient to meet the requirements of domain-specific tasks. In highly specialized applications such as pollution control—where semantic precision and regulatory logic are paramount—the integration of static domain knowledge embedding (fine-tuning) with dynamic knowledge augmentation (retrieval) is essential to improving LLM utility and reliability.

5.3. Engineering relevance and practical implications

Beyond technical performance improvements, the proposed RAG +

FT model demonstrates practical engineering relevance when applied to concrete regulatory compliance tasks. One representative scenario is clause-level question answering during pollution control compliance review, where practitioners must interpret detailed regulatory requirements and assess their applicability to specific engineering or management contexts. In a traditional workflow, this task typically requires manual searching across multiple policy documents, identification of relevant clauses, and careful interpretation of regulatory language, which is time-consuming and prone to omission or misinterpretation—especially for non-specialist stakeholders such as corporate compliance teams or field environmental personnel. Conventional information retrieval systems can assist in document lookup but provide limited support for clause-level reasoning or regulatory logic interpretation. To address these limitations, the proposed workflow can be summarized as follows: (1) a regulatory question is provided by the user; (2) relevant clauses are retrieved from the policy corpus; (3) a grounded answer is generated based on the retrieved clauses; and (4) the output is applied or reviewed in compliance assessment.

In contrast to the traditional approach, the RAG + FT framework retrieves relevant regulatory clauses and generates grounded responses that explicitly align with the original policy text. As evidenced by the expert evaluation results, this workflow yields higher Compliance and semantic consistency compared with baseline models, indicating improved alignment with regulatory intent. By reducing the need for manual cross-referencing and interpretation, the proposed system offers a practical tool for supporting regulatory understanding in compliance assessment, engineering design review, and pollution control planning.

From a systems perspective, it should be noted that the RAG + FT framework incurs additional inference overhead compared with a standalone language model, primarily due to the retrieval step and the increased input context length. This overhead mainly arises from embedding-based similarity search and clause retrieval, rather than from the generation process itself. In practical deployments, these components can be optimized through indexing, caching, or offline preprocessing. Given the substantial gains in Compliance and expert-evaluated reliability observed in this study, the additional computational cost may be acceptable for compliance-critical regulatory applications, where correctness and grounding are prioritized over minimal response latency.

5.4. Contribution to open research and community reuse

Beyond methodological contributions, this study offers the first openly available, structured knowledge base and QA dataset specifically focused on Chinese pollution control policies and technical guidelines. All resources — including pre-processed regulatory texts, fine-tuning corpora, and the Dify-based knowledge base construction workflow — are hosted on GitHub. These resources are designed with modularity and reusability in mind, enabling researchers and practitioners to:(1) Reproduce and validate the experiments presented in this paper.(2) Extend the dataset with new regulatory documents or domain-specific corpora.(3) Integrate the knowledge base into custom RAG pipelines for downstream applications such as regulatory Q&A, compliance checking, and decision support.(4) By releasing these assets under an open-source license, this work aims to lower the barrier for future research in intelligent environmental governance and promote collaborative advancements in the field.

5.5. Limitations and future work

Despite the promising outcomes, this study has several limitations that point toward directions for future research and improvement.

First, the scale and diversity of the dataset used in this study were limited. The training and evaluation primarily relied on Chinese regulatory texts and question–answer pairs related to pollution control, with a relatively small test set of 100 QA pairs. This limitation may constrain the model's generalization capacity and the statistical robustness of the results. Future work will focus on expanding the size and diversity of the test dataset by incorporating regulatory texts from multiple industrial sectors, various pollution media (e.g., soil, solid waste), and regulations from different temporal periods. This expansion will improve the model's robustness, representativeness, and external validity.

Second, the cross-lingual, cross-jurisdiction, and real-world applicability of the model remains to be validated. The current framework was primarily built and evaluated on curated Chinese-language regulatory texts, without accounting for structural differences, terminological variation, and document quality issues in other legal systems or deployment settings. In practice, environmental regulations are updated over time and may be recorded in incomplete or noisy formats, and different stakeholder groups (e.g., enterprises, regulators, consultants) may pose heterogeneous information needs. Future research should therefore extend the approach to English, Japanese, and EU environmental regulations, while simultaneously investigating robustness mechanisms such as multilingual knowledge representation, regulatory version tracking and corpus maintenance, retrieval-quality filtering, and human-in-the-loop review for high-stakes compliance decisions, with the goal of building a globally applicable and practically reliable intelligent policy interpretation system.

Third, while the expert evaluation helped compensate for the limitations of automated metrics, the number of experts and the diversity of their backgrounds were limited, and the evaluation was restricted to static QA tasks. Future work could broaden the expert pool and introduce interactive evaluation protocols and human–AI collaborative annotation approaches to improve the objectivity, representativeness, and ecological validity of model assessments, and to operationalize human-in-the-loop oversight as a safeguard for the accuracy, transparency, and compliance of AI-assisted regulatory interpretations.

Fourth, the present study does not explicitly analyze how domain-specific fine-tuning and retrieval augmentation reshape the internal semantic embedding space; the reported cosine similarity gains are measured in an external evaluation embedding space at the output level. Future work will investigate representation-space dynamics (e.g., concept-wise clustering and distance patterns before and after fine-tuning) and compare different domain adaptation strategies to better understand how LLMs encode pollution-control concepts and regulatory nuances.

Fifth, the present study does not include an explicit uncertainty estimation mechanism for the RAG + FT model; all reported variances are aggregate measures across folds or raters rather than calibrated prediction-level uncertainty. Future work will investigate uncertainty-aware designs, such as confidence-calibrated answering or abstention mechanisms for low-confidence regulatory QA.

In addition, the retrieval mechanism itself is kept fixed in the present study: all RAG and RAG + FT results are based on a hybrid dense–sparse retrieval pipeline with cross-encoder reranking. We do not perform a systematic ablation of alternative retrieval strategies (e.g., dense-only vs. sparse-only retrieval or different reranking schemes), so the individual contribution of each retrieval component to the overall performance remains to be quantified. Future work will compare alternative retrievers and weighting schemes to better understand their impact on regulatory QA performance and efficiency.

Finally, although this study focuses primarily on Chinese pollution control regulations, the RAG + FT architecture is inherently modular: the retrieval knowledge base, QA supervision data, and terminology resources are decoupled from the generative module. This modularity suggests that the framework can be adapted to other environmental regulatory domains—such as electronic-waste regulation or hazardous-materials management—by substituting the domain corpus and constructing corresponding QA supervision without redesigning the core inference pipeline. Future work will further examine such cross-domain extensions and evaluate the extent to which retrieval-grounded fine-tuning can generalize across regulatory subfields with heterogeneous

terminology structures and policy formulations. However, the present experiments are restricted to Chinese pollution control guidelines and do not establish validated performance in non-environmental regulatory domains such as healthcare or finance; extending the framework to such domains will require new domain-specific corpora, QA supervision, and dedicated evaluation.

6. Conclusion

This study presents a hybrid large language model framework that integrates Retrieval-Augmented Generation (RAG) with domain-specific fine-tuning (FT) to enhance the interpretation of complex pollution control guidelines. The results demonstrate that the hybrid approach effectively combines factual retrieval and contextual adaptation, achieving superior correctness, compliance, and clarity compared with single-strategy models. Beyond methodological advancement, the framework provides a practical pathway for intelligent environmental policy interpretation, enabling more efficient regulatory comprehension and decision support. The open release of structured datasets and workflows further promotes reproducibility and cross-domain extension. Overall, this work establishes a scalable foundation for data-driven, knowledge-enhanced environmental governance and offers new directions for the application of trustworthy AI in scientific policy management.

CRediT authorship contribution statement

Zhuhan Zhou: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Tingting Jiang:** Writing – review & editing, Supervision, Methodology. **Qiang Jin:** Writing – review & editing, Resources, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2026.128718. Most of the resources and outputs generated in this study have been made publicly available via GitHub. These include: official pollution prevention and control technical documents and guidelines released by the Ministry of Ecology and Environment of the People's Republic of China, the fine-tuning dataset, the fine-tuned model, evaluation datasets along with code, and the Dify-based text segmentation workflow. All data and models can be accessed through the following link: https://github.com/18756619952/Pollution-prevention-and-control-technology.

Data availability

Data will be made available on request.

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